

What Slows You Down

What Slows You Down: Classifying Delay Sources on Kathmandu's Urban Corridors Using Real-Time GPS and Accelerometer Observations

Research Questions

RQ1: What is the composition of travel delay by cause on three selected urban corridors in Kathmandu, and how does this composition vary by corridor and time of day?

RQ2: After controlling for intersection proximity, bus stop density, and market presence, does segment-level road surface roughness — measured via smartphone accelerometer — independently predict the frequency of vehicle slowdowns?

Objectives

Objective 1: To detect, classify, and rank the causes of vehicle delay events on three urban corridors using GPS speed profiles, accelerometer data, and real-time observer annotation from a moving vehicle.

Objective 2: To determine whether road surface roughness is a statistically significant independent predictor of segment-level slowdown frequency after accounting for other corridor features.

Methodology

Platform: Researcher’s own vehicle, driven along urban corridors at morning peak, evening peak, and off-peak periods.

Data collected per trip: GPS trace (1 Hz), accelerometer trace (50 Hz, Z-axis), voice-annotated cause codes at each slowdown, infrastructure checklist per segment.

Delay event: GPS speed below 5 km/hr sustained for more than 20 seconds.

Eight cause categories:

Code	Category	How Detected
S	Signal delay	Visual + known intersection location
Q	Unsignalized intersection queue	Visual
B	Bus / PT vehicle interference	Visual (bus stopped for boarding)
P	Pedestrian crossing	Visual
R	Road surface roughness	Visual + accelerometer Z-axis spike
F	Side friction	Visual (parking maneuver, loading, rickshaw)
T	Turning vehicle	

Code	Category	How Detected
E	Market / encroachment	Visual (U-turn, side road entry)
		Visual (vendor stalls on carriageway)

Sampling: 3 corridors $\times$ 3 time bands $\times$ 5 rides = 45 trips. Each corridor divided into 50-meter segments, observed $\sim$ 15 times across all passes.

Roughness: RMS of Z-axis accelerometer variance per 50m segment, averaged across passes, validated against visual observations.

Analysis:

Question	Method
RQ1	Descriptive: ranked cause lists, GIS hotspots
RQ2	Negative binomial mixed-effects regression: slowdown count $\sim$ roughness + signal distance + bus stop density + market presence + log(offset); random intercept per segment
Reliability	Intra-rater: 10% of trips video-recorded and re-coded by same observer after 2-week gap; Cohen's kappa between live and replay annotation
Power	Monte Carlo simulation, minimum detectable effect at 80%

Equipment: One smartphone.

Expected Outcomes

1. Per-corridor ranked delay composition — proportion of stopped time by each cause per corridor and time band.
2. GIS hotspot map of slowdown events, color-coded by dominant cause per 50m segment.
3. Regression results: effect size, confidence interval, and significance of roughness as a delay predictor after controls.
4. First smartphone-derived urban roughness dataset for Kathmandu corridors.
5. Per-corridor intervention recommendations matched to dominant cause: signal coordination, bus stop redesign, road resurfacing, or enforcement.